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# Physical foundation of the fluid particle dynamics method for colloid dynamics simulation

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Colloid dynamics is significantly influenced by many-body hydrodynamic interactions mediated by a suspending fluid. However, theoretical and numerical treatments of such interactions are extremely difficult. To overcome this situation, we developed a fluid particle dynamics (FPD) method [H. Tanaka and T. Araki, *Phys. Rev. Lett.*, 2000, **35**, 3523], which is based on two key approximations: (i) a colloidal particle is treated as a highly viscous particle and (ii) the viscosity profile is described by a smooth interfacial profile function. Approximation (i) makes our method free from the solid–fluid boundary condition, significantly simplifying the treatment of many-body hydrodynamic interactions while satisfying the incompressible condition without the Stokes approximation. Approximation (ii) allows us to incorporate an extra degree of freedom in a fluid, e.g., orientational order and concentration, as an additional field variable. Here, we consider two fundamental problems associated with these approximations. One is the introduction of thermal noise and the other is the incorporation of coupling of the colloid surface with an order parameter introduced into a fluid component, which is crucial when considering colloidal particles suspended in a complex fluid. Here, we show that our FPD method makes it possible to simulate colloid dynamics properly while including full hydrodynamic interactions, inertia effects, incompressibility, thermal noise, and additional degrees of freedom of a fluid, which may be relevant for wide applications in colloidal and soft matter science.

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## 1 Introduction

Colloidal suspensions are usually composed of solid (hard or soft) colloidal particles and (viscous or viscoelastic) liquid. The liquid component in a colloidal suspension has two important dynamical roles: one is to exert thermal force noise on colloid particles, which leads to their thermal Brownian motions, and another is to induce long-range hydrodynamic interactions between them.<sup>1,2</sup> So far, the diffusional transport of colloidal particles [role (i)] has mainly been considered as the key transport mechanism. However, since colloidal particles are suspended in a liquid component in reality, there should always be hydrodynamic interactions between colloids [role (ii)], which may play an essential role in many dynamic problems of colloidal suspensions. In ordinary problems, where only slow modes play important roles, we may assume incompressibility for a suspending liquid. Under this condition, the motion of colloids with compressible nature is not allowed, and only that of transverse nature is allowed. Such a constraint on the motion of colloids leads to a dynamical correlation between them, leading to a large impact on structural formation and its kinetic pathway of soft matter including a liquid component.<sup>3–8</sup> Thus, treating many-body

hydrodynamic interactions among colloidal particles is crucial for simulating the dynamical behavior of colloidal suspensions.

If we explicitly treat the solid–fluid boundaries of colloid particles, then enormous calculations are required for simulations of many-body problems due to the intrinsically long-range nature of hydrodynamic interactions. To overcome this difficulty, a number of numerical schemes have been developed: Stokesian Dynamics (SD),<sup>9,10</sup> Lattice Boltzmann (LB),<sup>11–16</sup> Dissipative Particle Dynamics (DPD),<sup>17</sup> Stochastic Rotational Dynamics (SRD),<sup>18–21</sup> Fluid Particle Dynamics (FPD)<sup>4,6,8</sup> and the Smoothed Profile Method (SPM).<sup>22–24</sup> A common feature among these simulation methods is that small, fast solvent molecules are treated in a coarse-grained manner. In such simulations, coupled dynamics of relevant variables, particle positions, flow field and order parameter(s) describing a complex fluid are numerically solved. Each simulation method has merits and demerits. For example, the incompressibility condition, which is usually satisfied for slow dynamics of colloidal suspensions, can be slightly violated in the LB and SRD methods. The SD method, by its nature, cannot treat nonlinear effects arising from the convective acceleration, which are important at a high Reynolds number, although such effects are not so important in most colloid problems. The incorporation of thermal noise is also not always so straightforward.<sup>25,26</sup> In many methods, it is also not easy to incorporate the internal degrees of freedom into a fluid component, which is necessary when we consider dynamics of

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charged colloids and colloids immersed in a complex fluid having an internal order parameter (e.g., superfluidity in a quantum liquid, density field in a fluid near a gas–liquid phase transition, concentration field near a critical point of a binary liquid mixture, and orientational order in liquid crystals). We may also want to simulate a liquid immersed in a viscoelastic fluid. As will be shown below, our FPD method can be almost free from most of these difficult problems. However, we need careful consideration and checks to validate a method incorporating each of these effects.

When we consider the dynamics of colloids, it is crucial to introduce the thermal force noise exerted on the solid colloid surface, which leads to thermal Brownian motion of colloidal particles. In FPD, we can apply a force noise to the center of mass of a particle, as done for SPM.<sup>27</sup> However, this way of introducing thermal noise only to solid particles has a problem, since the noise is not introduced in a manner consistent with the expected thermal fluctuation in a fluid. For example, in ref. 24, the temperature is determined by comparing the resulting diffusive motion of colloids using the fluctuation–dissipation theorem. However, since thermal fluctuations at different points in the fluid are coupled with each other through the Navier–Stokes equation, it is difficult to reproduce the resulting hydrodynamic memory effects properly by applying stochastic forces only at certain points; more precisely, it is hard to constitute such a random number sequence. In fact, in their paper, a disagreement of the velocity auto-correlation function between theory and numerical results was reported. Furthermore, thermal noise induces not only translation but also rotation of a colloid, but incorporating thermal noise to both degrees of freedom in a consistent manner is difficult in these types of methods.

In our FPD method, we can get rid of this difficulty as long as we are interested in the slow time regime where sound modes can be ignored. In our method, it is physically natural to exert thermal stress noise directly on the fluid including that inside a colloid, which is a viscous fluid particle in FPD. Since the fluctuation–dissipation relation is guaranteed in this scheme, this method is free from the above-mentioned problems, which is one of the merits of the FPD method. We will show this in this paper.

Colloids are often immersed in a complex fluid having an extra degree of freedom besides the momentum degree of freedom. When the colloidal surface is coupled with the internal degrees of freedom of the surrounding liquid, this coupling may play a crucial role in the dynamics of colloids. Such couplings can be seen in the following examples. (1) When colloidal particles are suspended in a binary fluid mixture, there is generally a direct coupling between the colloid surface and the concentration field *via* surface wetting effects.<sup>28–30</sup> (2) When colloidal particles are immersed in liquid crystals, the colloidal surface can also be coupled with the nematic or smectic order parameter *via* surface anchoring effects.<sup>31–33</sup> (3) When colloids have surface charges, the charges are coupled with ions in the liquid and the velocity field in a complicated manner.<sup>34,35</sup> In all these examples, the interactions between colloids can no longer be described by the static free energy or the Hamiltonian. Interactions are dynamically modulated,

reflecting the spatial distribution of the order parameters (the concentration field, the director field, and the ion concentration field for the above (1), (2) and (3), respectively). This is a consequence of the fact that the interaction is a function of the order parameter, whose spatial distribution is disturbed significantly by hydrodynamic interactions. Among these problems, cases (1) and (2) are essentially different from case (3) in the sense that the colloidal surface is coupled with the order parameter just on the colloid surface for cases (1) and (2), whereas the coupling is non-local for case (3). For such cases of local coupling at the surface, we need to take special care when we treat a colloidal particle with a smooth interface function. We also discuss this problem in this paper.

The organization of this paper is as follows: in Section 2, we describe how to incorporate thermal noise into the FPD method. In Section 3, we describe how to incorporate local coupling of the colloid surface to an order parameter of a fluid into the FPD method. Finally, we provide a summary in Section 4.

## 2 Incorporation of thermal force noise into the FPD method

Incorporation of thermal force noise into coarse-graining simulation methods is not so straightforward in many cases.<sup>25,26</sup> For stochastic rotational dynamics, or multiple-collision dynamics,<sup>18,19</sup> the thermal noise is rather naturally incorporated. However, for example, it is rather recent that thermal force noises have been incorporated into the LB method with full statistical mechanical consistency.<sup>12,26</sup> Here, we explain how we can incorporate thermal force noise into the FPD method in detail. Since our FPD method is based on the Navier–Stokes equation for a fluid, we can naturally incorporate thermal stress noise while keeping full statistical mechanical consistency by satisfying the fluctuation–dissipation theorem.<sup>36–38</sup> However, we need to take some care concerning the short-time behavior and the effect of finite thickness of the interface.

### 2.1 The fundamentals of the FPD method

Here, we briefly explain the FPD method<sup>4,39</sup> and the physical concept behind it. A particle whose center of mass is located at  $\mathbf{r}_i$  is represented by a smooth viscosity change as

$$\eta(\mathbf{r}) = \eta_\ell + \sum_i^N (\eta_c - \eta_\ell) \phi_i(\mathbf{r}), \quad (1)$$

where  $\eta_\ell$  is the viscosity of the liquid in which colloids are suspended, and  $\eta_c$  is the viscosity inside a colloid particle. The summation is taken over all  $N$  particles.  $\phi_i$  represents particle  $i$  as

$$\phi_i(\mathbf{r}) = \frac{1}{2} \left\{ \tanh \left[ \frac{1}{\xi_i} (a_i - |\mathbf{r} - \mathbf{r}_i|) \right] + 1 \right\}, \quad (2)$$

where  $a_i$  is the radius of particle  $i$  and  $\xi_i$  is the thickness of the particle interface. The key points of this method are that (i) the particle rigidity is approximately sustained by a large viscosity difference,  $(\eta_c - \eta_\ell)/\eta_\ell \gg 1$ , and (ii) a rigid solid–liquid boundary is replaced by a smooth interface with thickness  $\xi_i$ . When there is no other gross (or hydrodynamic) variable than

the velocity field  $\mathbf{v}(\mathbf{r})$ , the dynamics of colloidal particles with hydrodynamic interactions can be approximately described by solving the Navier–Stokes equation with eqn (1) and (2). It is worth noting that with decreasing  $\xi_i$  and increasing  $\eta_c$ , the method is expected to become asymptotically exact.<sup>4,6,40</sup>

First, we consider the simplest case, where colloid particles are immersed in a simple liquid and they are interacting with a hard-core-like repulsive potential. Then, the equation of motion to be solved is

$$\rho \left( \frac{\partial}{\partial t} + \mathbf{v} \cdot \nabla \right) \mathbf{v} = \mathbf{f} - \nabla \cdot [\bar{\mathbf{\Pi}} - \bar{\boldsymbol{\sigma}}] \quad (3)$$

with

$$\bar{\mathbf{\Pi}} = p\bar{\mathbf{I}} - \eta(\nabla \mathbf{v}^\dagger + \nabla \mathbf{v}),$$

where  $\bar{\mathbf{I}}$  is the unit tensor. In the above equation,  $\rho$  is the mass density, and we assume that the density of the liquid is the same as that of the particles.  $\mathbf{v}(\mathbf{r})$  is the velocity field, and the pressure  $p$  is determined to satisfy the incompressibility condition  $\nabla \cdot \mathbf{v} = 0$ .  $\bar{\boldsymbol{\sigma}}$  is the random stress noise, which we will discuss in detail in Section 2.2. In the above,  $\mathbf{f}(\mathbf{r})$  is the interparticle force density, which is obtained by distributing the force  $\mathbf{F}_i$  and torque  $\mathbf{N}_i$  acting on the center of mass of particle  $i$  to the lattice with the weight of  $\phi_i$ . Here, we decompose the force density to the translational and rotational components:  $\mathbf{f}(\mathbf{r}) = \mathbf{f}^T(\mathbf{r}) + \mathbf{f}^R(\mathbf{r})$ . The first and second terms of the right-hand side are the force density corresponding to  $\mathbf{F}_i$  and  $\mathbf{N}_i$ , respectively, and expressed as

$$\begin{aligned} \mathbf{f}^T(\mathbf{r}) &= \sum_i^N \phi_i(\mathbf{r}) \mathbf{F}_i / \int d\mathbf{r} \phi_i(\mathbf{r}), \\ \mathbf{f}^R(\mathbf{r}) &= \sum_i^N \phi_i(\mathbf{r}) \mathbf{r}_{oi} \times (\mathbf{N}_i \times \mathbf{r}_{oi}) / \int d\mathbf{r} \phi_i(\mathbf{r}) |\hat{\mathbf{N}}_i \times \mathbf{r}_{oi}|^2, \end{aligned}$$

where the rotation center of particle  $i$  is located at its center-of-mass,  $\mathbf{r}_{oi} = \mathbf{r} - \mathbf{r}_i$ , and  $\hat{\mathbf{N}}_i$  is the unit vector along  $\mathbf{N}_i$ . In our FPD method, the particle rigidity is approximately expressed by introducing the smooth viscosity profile,  $\eta(\mathbf{r})$ . The approximation is better for a larger viscosity ratio  $\eta_c/\eta_\ell$  and a smaller  $\xi_i/a_i$ . By multiplying both sides of eqn (3) by  $\phi_i(\mathbf{r})$  and then performing its spatial integration, we can straightforwardly obtain an approximate equation of the translational motion of particle  $i$ :

$$M_i \frac{d\mathbf{V}_i}{dt} = \mathbf{F}_i + \mathbf{K}_i^T, \quad (4)$$

where  $M_i = \lambda_i^T \rho \int d\mathbf{r} \phi_i(\mathbf{r})$  and  $\mathbf{V}_i = \int d\mathbf{r} \mathbf{v} \phi_i(\mathbf{r}) / \int d\mathbf{r} \phi_i(\mathbf{r})$  are the mass and the velocity of particle  $i$ , respectively. Here, we note that the time derivative of the particle velocity can be approximated as

$$d\mathbf{V}_i/dt \cong \int d\mathbf{r} (\partial/\partial t + \mathbf{v} \cdot \nabla) \mathbf{v} \phi_i(\mathbf{r}) / \int d\mathbf{r} \phi_i(\mathbf{r}). \quad (5)$$

In the above equation,  $\lambda_i^T$  is a correction factor resulting from the introduction of the smooth interface with the finite thickness  $\xi$  (see also the discussion in Section 2.3) and given by

$$\lambda_i^T = \int d\mathbf{r} \phi_i(\mathbf{r}) / \int d\mathbf{r} \phi_i^2(\mathbf{r}).$$

This correction factor not only makes the force acting on undeformable fluid particle  $i$  equal to  $\mathbf{F}_i$ , but also guarantees that the colloidal temperature is the same as the fluid temperatures (see Section 2.2). In the sharp-interface limit  $\xi_i/a_i \rightarrow 0$ ,  $\lambda_i^T$  tends to  $\rightarrow 1$ . The second term of the right-hand side of eqn (4) is the force exerted by the fluid, which is given by

$$\mathbf{K}_i^T = -\lambda_i^T \int d\mathbf{r} \phi_i \nabla \cdot [\bar{\mathbf{\Pi}} - \bar{\boldsymbol{\sigma}}] \cong -\lambda_i^T \int dS_i \hat{\mathbf{n}}_i \cdot [\bar{\mathbf{\Pi}} - \bar{\boldsymbol{\sigma}}].$$

Here, we use the following approximate relation

$$\int d\mathbf{r} \nabla \phi_i \cdot \bar{\mathbf{Q}} \cong - \int_{S_i} \hat{\mathbf{n}}_i dS_i \cdot \bar{\mathbf{Q}},$$

for an arbitrary tensor  $\bar{\mathbf{Q}}(\mathbf{r})$ , where  $S_i$  is the surface of particle  $i$  and  $\hat{\mathbf{n}}_i$  is the unit outward normal vector to  $S_i$ .

In practical numerical calculations, the on-lattice velocity field,  $\mathbf{v}(\mathbf{r}, t + \Delta t)$ , is evaluated from the physical quantities at time  $t$  by eqn (3). Then, we move particle  $i$  off-lattice as a rigid body by  $\mathbf{r}_i(t + \Delta t) = \mathbf{r}_i(t) + \Delta t \mathbf{V}_i(t + \Delta t)$ , where  $\Delta t$  is the time increment of the numerical integration.

Similarly, the equation of rotational motion of particle  $i$  can be obtained by operating  $\phi_i(\mathbf{r}) \mathbf{r}_{oi} \times$  on both sides of eqn (3) and then performing its spatial integration:

$$I_i \frac{d\boldsymbol{\Omega}_i}{dt} = \mathbf{N}_i + \mathbf{K}_i^R, \quad (6)$$

where  $I_i = \lambda_i^R \rho \int d\mathbf{r} |\hat{\mathbf{n}} \times \mathbf{r}_{oi}|^2 \phi_i(\mathbf{r})$  ( $\hat{\mathbf{n}}$  being a unit vector along the rotational axis) and  $\boldsymbol{\Omega}_i = \int d\mathbf{r} \phi_i(\mathbf{r}) (\mathbf{r}_{oi} \times \mathbf{v}) / \int d\mathbf{r} |\hat{\mathbf{n}} \times \mathbf{r}_{oi}|^2 \phi_i(\mathbf{r})$  are the moment of inertia and the angular velocity of particle  $i$ , respectively.  $\lambda_i^R$  is the correction factor regarding rotational motion, whose role is the same as  $\lambda_i^T$ , and it is given as

$$\lambda_i^R = \int d\mathbf{r} |\hat{\mathbf{n}} \times \mathbf{r}_{oi}|^2 \phi_i(\mathbf{r}) / \int d\mathbf{r} |\hat{\mathbf{n}} \times \mathbf{r}_{oi}|^2 \phi_i^2(\mathbf{r}).$$

The last term in eqn (6) is the torque exerted by the fluid, which can be approximated in the same way as the case of translational motion:

$$\begin{aligned} \mathbf{K}_i^R &= -\lambda_i^R \int d\mathbf{r} \phi_i \mathbf{r}_{oi} \times \nabla \cdot [\bar{\mathbf{\Pi}} - \bar{\boldsymbol{\sigma}}] \\ &\cong -\lambda_i^R \int dS_i \hat{\mathbf{n}}_i \cdot (\mathbf{r}_{oi} \times [\bar{\mathbf{\Pi}} - \bar{\boldsymbol{\sigma}}]) \end{aligned}$$

## 2.2 Introduction of thermal stress noise

In the FPD method, there are two ways to incorporate thermal noise. One is to introduce the thermal force noise to the center of mass of a particle as in Brownian dynamics simulation. Since the particle is treated as a solid particle for the motion of its center of mass, this is certainly a candidate. As mentioned in Introduction, it is difficult with this method to reproduce hydrodynamic memory that is inherent in Navier–Stokes equation. Furthermore, the rotational Brownian motion is not naturally introduced. The other method is to introduce the thermal stress noise directly into the fluid to satisfy the fluctuation–dissipation theorem with the nonlocal viscosity  $\eta(\mathbf{r})$ .<sup>8</sup> This should properly reproduce both translational and rotational Brownian motions

with full statistical mechanical consistency.<sup>36–38</sup> This theoretical framework is widely known as fluctuating hydrodynamics. Thus, we introduce the random stress noise,  $\vec{\sigma}(\mathbf{r})$ , which satisfies the fluctuation–dissipation relation:<sup>41</sup>

$$\langle \sigma_{ij}(\mathbf{r}, t) \sigma_{mn}(\mathbf{r}', t') \rangle = 2k_B T \eta(\mathbf{r}) (\delta_{im} \delta_{jn} + \delta_{in} \delta_{jm} - \frac{2}{3} \delta_{ij} \delta_{nm}) \delta(\mathbf{r} - \mathbf{r}') \delta(t - t').$$

The variance of the particle velocity at equilibrium is given by

$$\begin{aligned} \langle |\mathbf{V}_i|^2 \rangle &= \int d\mathbf{r} d\mathbf{r}' \phi_i(\mathbf{r}) \phi_i(\mathbf{r}') \langle \mathbf{v}(\mathbf{r}) \cdot \mathbf{v}(\mathbf{r}') \rangle / \left[ \int d\mathbf{r} \phi_i(\mathbf{r}) \right]^2 \\ &= 2k_B T / M_i. \end{aligned}$$

Here, we make use of the relation  $\langle \mathbf{v}(\mathbf{r}) \cdot \mathbf{v}(\mathbf{r}') \rangle = (2k_B T / \rho) \delta(\mathbf{r} - \mathbf{r}')$ . The relation  $\langle |\mathbf{V}_i|^2 \rangle = 2k_B T / M_i$  also represents the equipartition theorem for the kinetic energy with respect to the translational motion of the particle:  $M_{\text{eff}} \langle |\mathbf{V}_i|^2 \rangle / 2 = 3k_B T / 2$ , where  $M_{\text{eff}} = 3M_i / 2$  is the effective mass.<sup>41</sup> Regarding the angular velocity, we also obtain the equipartition theorem:  $\langle |\mathbf{\Omega}_i|^2 \rangle = 3k_B T / I_i$ .

### 2.3 On the correction factor originating from the finite interface thickness

Here, it may be worth commenting on the correction factor introduced above. Our basic set of kinetic equations exactly satisfy the fluctuation–dissipation relation. In our method, however, we need to cut out a high-viscosity region represented by our shape function  $\phi_i$  and regard it as a solid particle (the transformation from a fluid particle to a solid one). This process inevitably suffers from an artificial effect coming from the finite thickness of the interface. This inevitably leads to the necessity to introduce the correction factor. However, this effect must disappear in the sharp interface limit:  $\xi_i / a_i \rightarrow 0$ . What is important is that before the transformation from a fluid particle to a solid one, our FPD method satisfies both thermodynamic and mechanical principles in an exact manner and even after the transformation, this also holds with the correction factors (see the next section).

## 2.4 Results

**2.4.1 Setting of numerical simulations.** In this section, we check the validity of our simulation method incorporating thermal fluctuation effects by examining the Brownian motion of particles and fluid motion at a low volume fraction of colloids  $\Phi$ . Hereafter, we consider a monodisperse system and thus omit the particle index  $i$ . In our simulation, the units of length  $\ell$  and time  $\tau$  are related as  $\tau = \ell^2 / (\eta_c / \rho)$ , which sets both the scaled density and viscosity of the fluid region to unity. This  $\tau$  is a time required for the fluid momentum to diffuse over a lattice size  $\ell$ . The units of stress and energy are  $\bar{\sigma} = \rho(\ell/\tau)^2$  and  $\bar{\epsilon} = \bar{\sigma}\ell^3$ , respectively. Furthermore, we set  $\eta_c / \eta_c = 50$ ,  $\Delta t = 0.025$  and  $\xi = \ell = 1$ , and  $a = 3.2$ . The simulation box used was  $L^3 = 256^3$ . To avoid cumbersome expressions, we will use the same characters for the scaled variables below. We solve the equation of motion [eqn (3)] by using the Marker-and-Cell (MAC) method with a staggered lattice under periodic boundary conditions.

We put thermal noises on each lattice point, and thus a particle cannot be regarded as solid at short time. However, the higher viscosity inside a particle and the resulting quicker momentum diffusion lead to a quicker decay of the velocity fluctuations inside than outside: the velocity field inside a particle is homogenized and a particle can be regarded as solid for  $\sqrt{(\Delta r)^2} / a \geq O(0.1)$ , i.e. for  $t/\tau_B \gtrsim 0.1$  in our case.

**2.4.2 Fluid motion: velocity correlation.** We first focus on the hydrodynamic motion of a fluid and measure the velocity auto-correlation function (VAF). Here, we put 100 colloidal particles in the simulation box and apply the thermal noise such that  $k_B T = 14.3$ . The volume fraction  $\Phi = 4\pi a^3 N / 3L^3$  is estimated as  $\sim 0.001$ . This situation can be regarded as a dilute limit, where the translational and rotational VAF is expected to obey the following relation:

$$\langle \mathbf{V}(t) \cdot \mathbf{V}(0) \rangle = \frac{3k_B T}{M_{\text{eff}}} \psi^T \left( \frac{t}{\tau_a} \right). \quad (7)$$

$$\langle \mathbf{\Omega}(t) \cdot \mathbf{\Omega}(0) \rangle = \frac{3k_B T}{I} \psi^R \left( \frac{t}{\tau_a} \right). \quad (8)$$

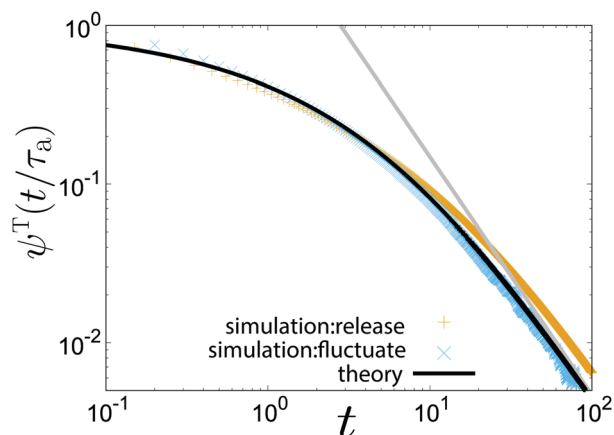
Here,  $\tau_a = a_h^2 / \nu_c$ , where  $a_h$  is the hydrodynamic radius ( $a_h = 1.04a$  and  $1.14a$  for translational and rotational motion respectively) and  $\nu_c = \eta_c / \rho$  is the momentum diffusion constant of the fluid.  $\psi^T$  and  $\psi^R$  are the memory functions including the hydrodynamic effect for translational and rotational motion, respectively. An expression of  $\psi^T$  and  $\psi^R$  is given by<sup>42,43</sup>

$$\psi^T(x) = \frac{\alpha_+ w(i\alpha_+ \sqrt{x}) - \alpha_- w(i\alpha_- \sqrt{x})}{(\alpha_+ - \alpha_-)}, \quad (9)$$

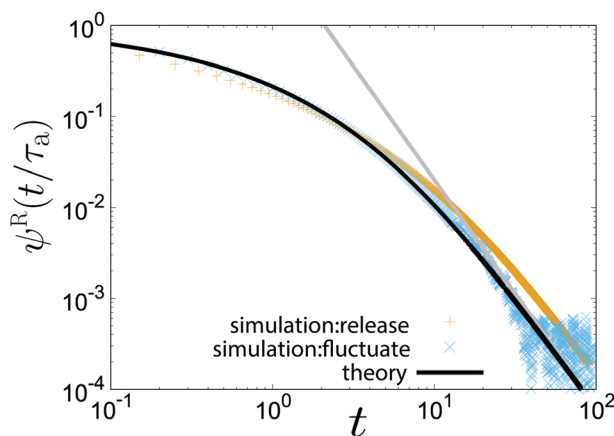
$$\begin{aligned} \psi^R(x) &= \frac{(1+z_1)z_1 w(-iz_1 \sqrt{x})}{(z_1 - z_2)(z_1 - z_3)} \\ &+ \frac{(1+z_2)z_2 w(-iz_2 \sqrt{x})}{(z_2 - z_3)(z_2 - z_1)} \\ &+ \frac{(1+z_3)z_3 w(-iz_3 \sqrt{x})}{(z_3 - z_1)(z_3 - z_2)}, \end{aligned} \quad (10)$$

where  $w$  is the complex complementary error function,  $\alpha_{\pm} = \frac{3}{2} \pm \frac{\sqrt{3}}{2}i$ , and  $z_i$  ( $i = 1, 2, 3$ ) are the solutions of the equation  $x^3 + 6x + 15 = 0$ . The asymptotic behaviors of  $\psi^T$  and  $\psi^R$  in the large  $x$  (or, long-time) limit are given by  $\psi^T(x) \sim \frac{1}{6\sqrt{\pi}} x^{-3/2}$  and  $\psi^R(x) \sim \frac{1}{60\sqrt{\pi}} x^{-5/2}$ , which are widely known as the long-time tails.

In Fig. 1 and 2, we plot the VAFs for translational and rotational motions calculated by the FPD method together with the above theoretical predictions, respectively. We can see very good agreements between them for both types of motion. In addition, we also checked whether our simulation method satisfies the fluctuation–dissipation theorem (FDT). To do so, we first put a particle in the simulation box and externally exert a constant drag force  $F_{\text{drag}} = 1$  or a constant torque  $N_{\text{drag}} = 1$  to drag or rotate the particle, respectively. After reaching the



**Fig. 1** Translational velocity auto-correlation function (TVAF). Blue symbols are the TVAF obtained by the FPD method for particles undergoing thermal Brownian motion at a dilute volume fraction ( $\sim 0.1\%$ ). Brown crosses are the TVAF obtained by the FPD method by studying the behavior after removing a drag force on a particle from the steady state. The black solid curve is the theoretical prediction (see eqn (9)). The gray line shows the asymptotic power-law decay function at a long-time limit:  $\psi^T(x) \sim (6\sqrt{\pi})^{-1} x^{-3/2}$ .



**Fig. 2** Rotational velocity auto-correlation function (RVAF). Blue symbols are the RVAF obtained by the FPD method for particles undergoing thermal Brownian motion at a dilute volume fraction ( $\sim 0.1\%$ ). Brown crosses are the RVAF obtained by the FPD method by studying the behavior after removing an external torque on a particle from the steady state. The black solid curve is the theoretical prediction (see eqn (11)). The gray line shows the asymptotic power-law decay function at a long-time limit:  $\psi^R(x) \sim (60\sqrt{\pi})^{-1} x^{-5/2}$ .

steady state, we remove the external force or the torque at  $t = 0$  and then monitor the decay of the translational velocity of the particle  $V(t)$  or that of the angular velocity  $\Omega(t)$ , respectively. In this situation, the following relations should be held:

$$\begin{aligned} \langle V(t) \cdot V(0) \rangle &= -\frac{3k_B T}{F_{\text{drag}}} \frac{dV_1(t)}{dt}, \\ \langle \Omega(t) \cdot \Omega(0) \rangle &= -\frac{3k_B T}{N_{\text{drag}}} \frac{d\Omega_1(t)}{dt}, \end{aligned}$$

The brown crosses in Fig. 1 and 2 are the VAFs measured using this procedure. The good agreements between the theoretical

predictions and the VAFs evaluated by the two different methods indicate that our method reproduces the FDT well, including the memory effect of hydrodynamic origin.

**2.4.3 Brownian motion of a particle.** Next, we examine the thermal Brownian motion of a colloid particle. First we focus on the translational motion of the particle. From eqn (9) and the following identity of the VAF,  $\langle V(t) \cdot V(0) \rangle = \frac{1}{2} \frac{d^2}{dt^2} \langle |\Delta r|^2 \rangle$ , where  $\Delta r(t) = r(t) - r(0)$  is the displacement vector of the particle, we obtain the following expression for the temporal evolution of the mean square displacement (MSD) of a free Brownian particle including hydrodynamic memory effects:

$$\frac{\langle |\Delta r(t)|^2 \rangle}{a^2} = \frac{t}{\tau_B} h\left(\frac{t}{\tau_a}\right). \quad (11)$$

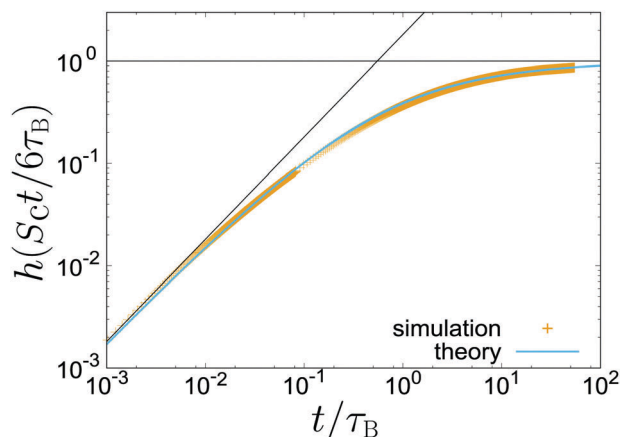
Here,  $\tau_B$  is the Brownian time  $\tau_B = a_h^2/6D^T$ , where  $D^T$  is the translational diffusion constant of a free particle given by  $D^T = k_B T / (\lambda^T 6\pi\eta a_h)$ , and the function  $h$  can be given by<sup>42</sup>

$$\begin{aligned} h(x) &= 1 - \frac{2}{\sqrt{\pi}} x^{-1/2} + \frac{2}{3} x^{-1} + \frac{3}{\alpha_+ - \alpha_-} x^{-1} \\ &\times \{ \alpha_+^{-3} w(i\alpha_+ \sqrt{x}) - \alpha_-^{-3} w(i\alpha_- \sqrt{x}) \}. \end{aligned} \quad (12)$$

The asymptotic behavior of  $h(x)$  is given by  $h(x) \rightarrow 3x/2$  ( $x \rightarrow 0$ ) and  $h(x) \rightarrow 1$  ( $x \rightarrow \infty$ ) for the ballistic and diffusive regimes, respectively. Thus, we can obtain from eqn (11) the following expressions for the short-time and long-time behaviors:  $\langle |\Delta r(t)|^2 \rangle = (3k_B T/M_{\text{eff}})t^2$  and  $\langle |\Delta r(t)|^2 \rangle = 6D^T t$ , respectively. The argument of  $h$  in eqn (11) can be rewritten as  $t/\tau_a = S_c t/6\tau_B$ , where  $S_c$  is called the “Schmidt number” and is defined as  $S_c = \nu_r/D^T$ .  $S_c$  is the ratio between the momentum and particle diffusion constants, and it determines the relation between the time unit of our simulation,  $\tau$ , and the crossover time from the ballistic to the diffusive motion of a particle.

In Fig. 3, we plot the function  $h$  obtained by analyzing the results of FPD simulations with the time unit  $\tau_B$  and compare it with eqn (12). We can see that our simulation method precisely reproduces Brownian motion from the ballistic to the diffusive regime. Here, we stress that in the time regime of  $t/\tau_B \gtrsim 1$ ,  $h$  is close to 1. This indicates that the Brownian motion in this simulation can be regarded to be almost in the over-damped regime. This situation is realized by choosing the strength of thermal fluctuation such that  $S_c$  is larger than 1 (in this particular simulation, we set  $S_c = 7.3$ ). This can be easily understood from the tilted straight line in Fig. 3,  $h = S_c t/4\tau_B$ : this line is shifted upward and downward upon decreasing and increasing the value of  $S_c$ , respectively. For simulations with  $S_c \ll 1$ , the ballistic motion becomes dominant even around the time regime  $t/\tau_B \sim 1$ . We note that the time unit of our simulation is set to the characteristic momentum diffusion time over  $\zeta$ , and thus increasing  $S_c$  requires a longer simulation time to reach the diffusive regime of the Brownian motion.

Next, we examine the rotational Brownian diffusion of a colloid particle. Here, we use a parameter  $\Delta u(t) = \frac{1}{2} \sum_{i=1,2,3} u_i(0) \times u_i(t)$  to measure the displacement of rotational motion of the particle,<sup>44</sup>



**Fig. 3** Crossover behavior from the ballistic to diffusive motion of a Brownian particle. We plot the function,  $h$ , given by eqn (11). The blue solid curve is the theoretical prediction (see eqn (12)), which can be approximated as  $h \sim S_c t / 4\tau_B$  in the short-time ballistic limit (shown by the black tilted line) whereas  $h \sim 1$  in the long-time diffusive limit (shown by the black horizontal line). Brown symbols are the results obtained by the FPD method for particles undergoing thermal Brownian motion at a dilute volume fraction ( $\sim 0.1\%$ ).

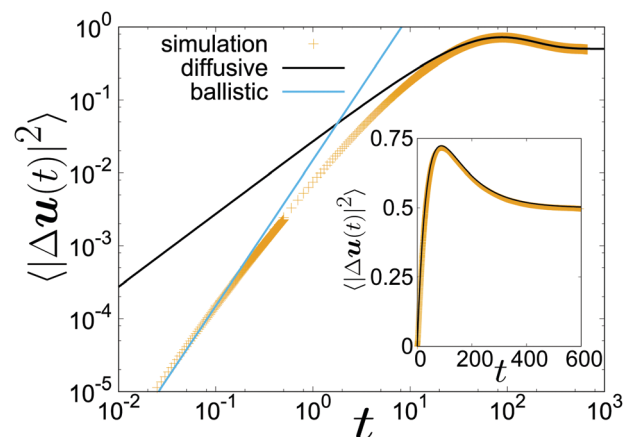
where  $\mathbf{u}_i(t)$  ( $i = 1, 2, 3$ ) are three unit vectors fixed on the particle that are orthogonal with each other. Time evolution of  $\mathbf{u}_i(t)$  is implemented by using a normalized quaternion (see, e.g. ref. 45). Denoting the axis of rotation of the particle from time 0 to  $t$  as  $\hat{\mathbf{n}}$  and the right-handed rotational angle as  $\theta$ , we can obtain  $\Delta\mathbf{u}(t) = \sin(\theta)\hat{\mathbf{n}}$ . Then, the “rotational mean square displacement” (RMSD) can be written as  $\langle |\Delta\mathbf{u}(t)|^2 \rangle = \langle \sin^2 \theta \rangle$ . In the diffusive regime, it can be calculated as  $\langle |\Delta\mathbf{u}(t)|^2 \rangle = \int_0^\pi \sin^2 \theta P(\theta) d\theta = 1/2 - (5e^{-6D^R t} - 3e^{-2D^R t})/4$ , where  $P(\theta)$  is the probability density of  $\theta$ :<sup>46</sup>

$$P(\theta) = \pi^{-1} \sum_{n=0}^{\infty} e^{-n(n+1)D^R t} (2n+1) \times \{\cos(n\theta) - \cos((n+1)\theta)\} \quad (13)$$

Here,  $D^R$  is the rotational diffusion constant:  $D^R = k_B T / (\lambda^R 8\pi\eta_s a_h^3)$ . In short time, RMSD can be approximated as  $\langle |\Delta\mathbf{u}(t)|^2 \rangle \sim \langle |\dot{\mathbf{u}}|^2 \rangle \sim \langle |\dot{\mathbf{u}}|^2 \rangle t^2 = 3k_B T^2 / I$ , which can be regarded as the ballistic regime. In Fig. 4, we plot the RMSD and we can see that our simulation method exhibits ballistic and diffusive behavior also for the rotational Brownian motion, following the theoretical predictions.

### 3 Incorporation of local coupling of the colloid surface to the order parameter of a fluid into the FPD method

Next, we consider application of the FPD method to colloids immersed in a complex fluid, which has an additional internal degree of freedom. In particular, we need to take special care to retain the thermodynamic consistency when there is local coupling of the colloid surface with a degree of freedom of a fluid. In our previous work, this point was not treated properly. In particular, it turned out<sup>30</sup> that this problem is serious when the interfacial force plays a dominant role in the dynamical behaviour,



**Fig. 4** The rotational mean-square displacement (RMSD) of a Brownian particle. Brown symbols are RMSD obtained by the FPD method for particles undergoing thermal Brownian motion at a dilute volume fraction ( $\sim 0.1\%$ ). The blue line and black curve are predictions for the short-time ballistic regime ( $\langle |\Delta\mathbf{u}(t)|^2 \rangle = 3k_B T^2 t^2 / I$ ) and for the long-time diffusive regime ( $\langle |\Delta\mathbf{u}(t)|^2 \rangle = 1/2 - (5e^{-6D^R t} - 3e^{-2D^R t})/4$ ).

as in a system of colloids immersed in a binary mixture. It may even lead to an unphysical result.<sup>29</sup> As a prototype example, we consider application of the FPD method to a colloid immersed in a binary mixture, in which the composition of the mixture is coupled to the colloid surface through surface wetting effects.

In the following, we ignore thermal noise for simplicity, although it is straightforward to incorporate concentration and stress noises into our method.

#### 3.1 FPD for colloids immersed in a liquid mixture

Here, we explain how to construct dynamic equations in the case where the concentration field  $\psi(\mathbf{r})$  and its surface energy at the solid–liquid interface become important. The method we will discuss below can be straightforwardly applied to a coupling of the colloid surface with other gross variables, such as the director field in liquid crystals. The free energy functional for  $\psi(\mathbf{r})$  and  $\mathbf{r}_i$  is given by

$$\mathcal{F}\{\psi, \mathbf{r}_i\} = F_b\{\psi\} + F_s\{\psi, \mathbf{r}_i\} + F_a\{\psi, \mathbf{r}_i\} + U(\mathbf{r}_i). \quad (14)$$

The first term is the coarse-grained free energy with the standard Ginzburg–Landau form,

$$F_b\{\psi\} = \int d\mathbf{r} \left[ f_0(\psi, T) + \frac{K_0}{2} |\nabla\psi|^2 \right], \quad (15)$$

where  $f_0(\psi, T)$  is the bulk free energy functional depending on  $\psi$  and temperature  $T$ :

$$f_0(\psi, T) = (1/2)r_0\varepsilon\psi^2 + (1/4)u_0\psi^4,$$

where  $\varepsilon$  is the reduced temperature and  $\varepsilon = (T - T_c)/T_c$  ( $T_c$  being the critical temperature of the binary mixture). Then, the coefficient  $K_0$  of the gradient term is a positive constant. The second term  $F_s$  represents the “surface” part written as

$$F_s\{\psi, \mathbf{r}_i\} = \sum_i^N \int d\mathbf{r} |\nabla\phi_i| f_s(\psi, T). \quad (16)$$

Here, the surface free energy  $f_s(\psi, T)$  depends on  $\psi$  and  $T$ , and we employ the following form:  $f_s = \gamma\psi$ , where  $\gamma$  is the strength of the surface field that depends on the details of the actual interaction between the colloid surface and the two species of the solvent.

We note that eqn (16) reduces to  $\sum_i^N \int d\mathbf{r} \delta(|\mathbf{r} - \mathbf{r}_i| - a) f_s(\psi, T)$  in the sharp-interface limit ( $\xi/a \rightarrow 0$ ). The third term  $F_a$  is the “artificial” thermodynamic potential,

$$F_a\{\psi, \mathbf{r}_i\} = \sum_i^N \int d\mathbf{r} \chi \phi_i \psi^2, \quad (17)$$

where  $\chi$  is a positive constant. This term is introduced to keep the  $\psi$  field inoperative inside the colloid, as it should be for a solid colloid particle. **This term prevents the concentration flux through the interface.** Its effectiveness can be confirmed by the fact that our result recovers the sharp interface solution except for the interface region (see Fig. 5).

Finally,  $U(\mathbf{r}_i) = \sum_{i>j} u(|\mathbf{r}_i - \mathbf{r}_j|)$  represents the direct interaction potential between particles, which is represented by  $u(r)$ .

Then, the dynamic equations can be described as follows. First, the velocity field obeys the Navier–Stokes equation,

$$\rho \left( \frac{\partial}{\partial t} + \mathbf{v} \cdot \nabla \right) \mathbf{v} = \mathbf{f}_{\text{rev}} - \nabla \cdot \vec{\sigma} \quad (18)$$

Here,  $\vec{\sigma} = p\vec{\mathbf{I}} - \eta(\nabla\mathbf{v}^\dagger + \nabla\mathbf{v})$  (note that here we define  $\sigma$  differently from Section 2), where  $\vec{\mathbf{I}}$  is the unit tensor, and the hydrostatic pressure  $p$  is determined to satisfy the incompressibility condition  $\nabla \cdot \mathbf{v} = 0$ . Then,  $\mathbf{f}_{\text{rev}}(\mathbf{r})$  is the reversible force density, which will be given below. Here, we assume for simplicity that the density of the liquid  $\rho$  is the same as that of the particles. The composition field  $\psi(\mathbf{r})$ , on the other hand, satisfies the continuity equation,

$$\frac{\partial}{\partial t} \psi = -\nabla \cdot \mathbf{j}_\psi. \quad (19)$$

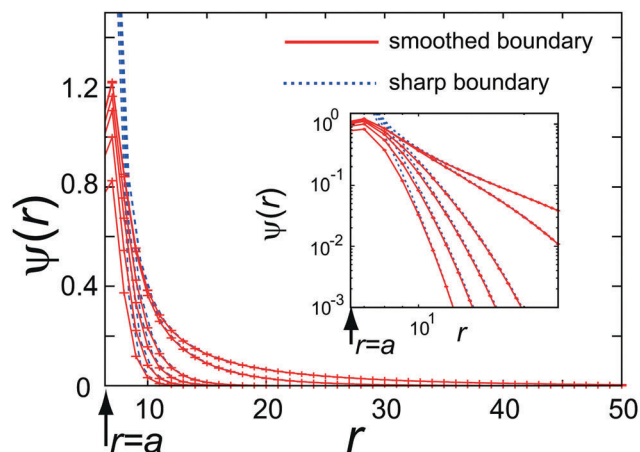


Fig. 5 The concentration profile around the particle for various values of reduced temperature  $\varepsilon$  ( $\varepsilon = 2, 1, 0.5, 0.25, 0.025, 0.0025$  from bottom to top curves) in both the sharp and smooth boundary cases. The log–log plot is shown in the inset.

Here,  $\mathbf{j}_\psi$  represents the flux of  $\psi$  given by

$$\mathbf{j}_\psi = \psi \mathbf{v} - L_\psi \nabla \frac{\delta \mathcal{F}}{\delta \psi}, \quad (20)$$

in which  $L_\psi$  is the kinetic coefficient for  $\psi$ . **Here, we set**

**$L_\psi = L_0 \left( 1 - \sum_i^N \phi_i \right)$ , where  $L_0$  is assumed to be a positive constant. This form is employed so that there is no diffusion flux inside the particle region ( $\phi_i = 1$ ). This choice of  $L_\psi$  and the inclusion of  $F_a$  ensure that the concentration field inside the particle region remains unchanged, i.e.  $\psi(\mathbf{r}, t) = 0$ , during the time integration.** Finally, the velocity of particle  $i$  is defined as the following average value:

$$\mathbf{V}_i = \frac{1}{\Omega_i} \int d\mathbf{r} \phi_i \mathbf{v}, \quad (21)$$

where  $\Omega_i = \int d\mathbf{r} \phi_i$ .

### 3.2 Derivation of the reversible force density

First, we derive the reversible force stemming from the concentration fluctuations and the particle motion for both the sharp and smooth colloid interfaces. Below, we neglect the dissipative effect for simplicity. Here, we estimate the reversible force density, according to the standard procedure.<sup>47</sup>

**3.2.1 The case of a sharp solid–liquid surface.** In the presence of a sharp surface, the free energy of the system is given by

$$\mathcal{F}\{\psi, \mathbf{r}_i\} = \int_{\hat{V}} d\mathbf{r} \left[ f_0(\psi) + \frac{K_0}{2} |\nabla \psi|^2 \right] + \sum_i \int_{\hat{S}_i} dS f_s(\psi) + U(\mathbf{r}_i), \quad (22)$$

where  $\int_{\hat{V}} d\mathbf{r}$  and  $\int_{\hat{S}_i} dS$  denote integration over the bulk and the surface of particle  $i$ , respectively. Due to the presence of the velocity field  $\mathbf{v}$ , the concentration changes during an infinitesimal time interval  $\delta t$  as  $\psi \rightarrow \psi' = \psi - \delta t \mathbf{v} \cdot \nabla \psi$  and the particle moves as  $\mathbf{r}_i \rightarrow \mathbf{r}_i' = \mathbf{r}_i + \delta t \mathbf{V}_i$ , where  $\mathbf{V}_i$  is the velocity of particle  $i$ , and it is identical to the  $\mathbf{v}$  at the surface of the particle ( $|\mathbf{r} - \mathbf{r}_i| = a$ ). The change in the free energy due to these changes is given by

$$\begin{aligned} \delta \mathcal{F}\{\psi, \mathbf{r}_i\} &= \mathcal{F}\{\psi', \mathbf{r}_i'\} - \mathcal{F}\{\psi, \mathbf{r}_i\} \\ &= \int_{\hat{V}} d\mathbf{r} \delta t \mathbf{v} \cdot \nabla \cdot \vec{\Pi} \\ &\quad + \sum_i \delta t \mathbf{V}_i \cdot \left( \int_{\hat{S}_i} dS \hat{\mathbf{n}}_i \cdot \vec{\Pi} + \frac{\partial U}{\partial \mathbf{r}_i} \right), \end{aligned} \quad (23)$$

where  $\vec{\Pi}$  is the osmotic pressure tensor given by eqn (27) (note that here we define  $\Pi$  differently from Section 2), and  $\hat{\mathbf{n}}_i$  is the unit vector normal to the surface of particle  $i$ . Eqn (23) represents the total work done on the fluid and particle, which is equal to the sign-inverted free energy difference without the dissipative effect. Thus, the force density acting on the bulk fluid and particle  $i$  is, respectively,  $-\nabla \cdot \vec{\Pi}$  and  $-\left( \int_{\hat{S}_i} dS \hat{\mathbf{n}}_i \cdot \vec{\Pi} + \partial U / \partial \mathbf{r}_i \right)$ . It should be noted that there is no contribution from the surface energy in eqn (23).

**3.2.2 The case of a smooth interface.** In the FPD method, we start from the free energy given by eqn (14). Due to the presence of the velocity field  $\mathbf{v}$ , the concentration changes during an infinitesimal time interval  $\delta t$  as  $\psi \rightarrow \psi' = \psi - \delta t \mathbf{v} \cdot \nabla \psi$  and the particle moves as  $\mathbf{r}_i \rightarrow \mathbf{r}'_i = \mathbf{r}_i + \delta t \int d\mathbf{r} \mathbf{v} \phi_i / \Omega_i$ . The change in the free energy due to these changes is given by

$$\begin{aligned} \delta \mathcal{F} \{\psi, \mathbf{r}_i\} &= \mathcal{F} \{\psi', \mathbf{r}'_i\} - \mathcal{F} \{\psi, \mathbf{r}_i\} \\ &= \int d\mathbf{r} \delta t \mathbf{v} \cdot \left( \psi \nabla \frac{\delta \mathcal{F}}{\delta \psi} + \sum_i \frac{\phi_i}{\Omega_i} \frac{\partial \mathcal{F}}{\partial \mathbf{r}_i} \right) \\ &= \int d\mathbf{r} \delta t \mathbf{v} \cdot \left\{ \psi \nabla \frac{\delta F_b}{\delta \psi} - \sum_i \left( |\nabla \phi_i| \nabla f_s - \frac{\phi_i}{\Omega_i} \frac{\partial F_s}{\partial \mathbf{r}_i} \right) \right. \\ &\quad \left. - \sum_i \left[ \phi_i \nabla (\chi \psi^2) - \frac{\phi_i}{\Omega_i} \frac{\partial F_a}{\partial \mathbf{r}_i} \right] + \sum_i \frac{\phi_i}{\Omega_i} \frac{\partial U}{\partial \mathbf{r}_i} \right\} \\ &= - \int d\mathbf{r} \delta t \mathbf{v} \cdot \mathbf{f}_{\text{rev}}. \end{aligned}$$

This is nothing but the definition of the reversible force density  $\mathbf{f}_{\text{rev}}$  acting on the fluid. Here, we use the relation  $\int d\mathbf{r} \mathbf{v} \cdot [\psi \nabla \delta(F_s + F_a) / \delta \psi] = - \int d\mathbf{r} \mathbf{v} \cdot \nabla \psi [\delta(F_s + F_a) / \delta \psi]$ . It should be noted that  $\delta F_s \{\psi, \mathbf{r}_i\}$  and  $\delta F_a \{\psi, \mathbf{r}_i\}$  vanish at the sharp interface and high viscosity limits ( $\xi/a \rightarrow 0$  and  $\eta_c \rightarrow \infty$ , respectively). For example,  $\delta F_s$  is expressed as

$$\begin{aligned} \delta F_s \{\psi, \mathbf{r}_i\} &= \int d\mathbf{r} \delta t \mathbf{v} \cdot \left( \psi \nabla \frac{\delta F_s}{\delta \psi} + \frac{\phi_i}{\Omega_i} \frac{\partial F_s}{\partial \mathbf{r}_i} \right) \\ &= \int d\mathbf{r} \delta t \mathbf{v} \cdot \left( -\nabla f_s |\nabla \phi_i| + \frac{\phi_i}{\Omega_i} \frac{\partial F_s}{\partial \mathbf{r}_i} \right) \\ &= \int d\mathbf{r} \delta t [|\nabla \phi_i| \nabla f_s \cdot (\mathbf{V}_i - \mathbf{v})], \end{aligned}$$

where  $\mathbf{V}_i = \int d\mathbf{r} \mathbf{v} \phi_i / \Omega_i$ , and we make use of the relation  $\nabla |\nabla \phi_i| = -(\partial/\partial \mathbf{r}_i) |\nabla \phi_i|$ . In the limits of  $\xi/a \rightarrow 0$  and  $\eta_c \rightarrow \infty$ ,  $\mathbf{v} \rightarrow \mathbf{V}_i$  for  $|\mathbf{r} - \mathbf{r}_i| \leq a$ , and thus  $\delta F_s \rightarrow 0$ . Similarly, we can show that  $\delta F_a \rightarrow 0$  in these limits. The volume integral of  $\mathbf{f}_{\text{rev}}$  over the region  $A_i$  ( $|\mathbf{r} - \mathbf{r}_i| \leq a$ ) gives the total force acting on it as follows:

$$\begin{aligned} \int_{A_i} d\mathbf{r} \mathbf{f}_{\text{rev}} &= - \left( \int_{S_i} d\mathbf{S} \hat{\mathbf{n}}_i \cdot \vec{\Pi} + \lambda_i^A \frac{\partial U}{\partial \mathbf{r}_i} \right) \\ &\quad - \left( - \int_{A_i} d\mathbf{r} |\nabla \phi_i| \nabla f_s + \lambda_i^A \frac{\partial F_s}{\partial \mathbf{r}_i} \right), \quad (24) \\ &\quad - \left( - \int_{A_i} d\mathbf{r} \phi_i \nabla \chi \psi^2 + \lambda_i^A \frac{\partial F_a}{\partial \mathbf{r}_i} \right), \end{aligned}$$

where  $\lambda_i^A = \int_{A_i} d\mathbf{r} \phi_i / \Omega_i$ , and  $S_i$  denotes the surface of  $A_i$ . Note that in the sharp interface limit, the second and third terms in eqn (24) vanish and  $\lambda_i^A = 1$ , so that  $\lim_{\xi/a \rightarrow 0} \int_{A_i} d\mathbf{r} \mathbf{f}_{\text{rev}} = - \left( \int_{S_i} d\mathbf{S} \hat{\mathbf{n}}_i \cdot \vec{\Pi} + \partial U / \partial \mathbf{r}_i \right)$ .

### 3.3 Expression of the reversible force density

From eqn (14)–(21), we can determine the form of the reversible force density as

$$\mathbf{f}_{\text{rev}} = -\psi \nabla \frac{\delta \mathcal{F}}{\delta \psi} - \sum_i^N \frac{\phi_i}{\Omega_i} \frac{\partial \mathcal{F}}{\partial \mathbf{r}_i}, \quad (25)$$

To clarify the contribution from each term of eqn (14), we rewrite eqn (25) as

$$\begin{aligned} \mathbf{f}_{\text{rev}} &= -\psi \nabla \frac{\delta F_b}{\delta \psi} - \sum_i^N \frac{\phi_i}{\Omega_i} \frac{\partial U}{\partial \mathbf{r}_i} \\ &\quad + \sum_i^N \left( \nabla f_s |\nabla \phi_i| - \frac{\phi_i}{\Omega_i} \frac{\partial F_s}{\partial \mathbf{r}_i} \right) \\ &\quad + \sum_i^N \left[ \nabla (\chi \psi^2) \phi_i - \frac{\phi_i}{\Omega_i} \frac{\partial F_a}{\partial \mathbf{r}_i} \right]. \quad (26) \end{aligned}$$

In eqn (26), the first term on the right-hand side represents the usual (bulk) osmotic force due to the chemical potential gradient, which can be expressed in the form of the divergence of the osmotic pressure tensor as<sup>47</sup>

$$-\psi \nabla \frac{\delta F_b}{\delta \psi} = -\nabla \cdot \vec{\Pi}$$

with

$$\vec{\Pi} = \left( \psi \frac{\delta F_b}{\delta \psi} - f_0 - \frac{K_0}{2} |\nabla \psi|^2 \right) \vec{\mathbf{I}} + K_0 \nabla \psi \nabla \psi. \quad (27)$$

On the other hand, the second and third terms on the right-hand side of eqn (26) originate from  $F_s$  and  $F_a$ , respectively, which vanish in the sharp-interface limit ( $\xi/a \rightarrow 0$ ). It should be mentioned that with eqn (26), the time derivative of the total free energy, which is the sum of  $F\{\psi\}$  and the kinetic energy  $\mathcal{F}_K\{\mathbf{v}\} = \int d\mathbf{r} \rho \mathbf{v}^2 / 2$ , is negative definite in the case without external forces, as it should be:

$$\frac{d}{dt} (\mathcal{F} + \mathcal{F}_K) = - \int d\mathbf{r} \left[ L \left( \nabla \frac{\delta \mathcal{F}}{\delta \psi} \right)^2 + \frac{\eta}{2} \sum_{m,n} (\nabla_m v_n + \nabla_n v_m)^2 \right] \leq 0.$$

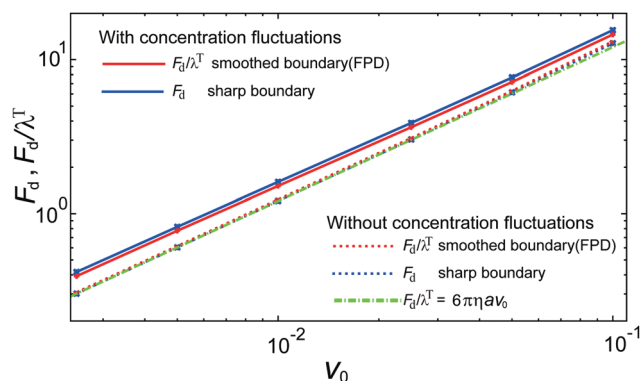
### 3.4 Numerical results

In practical numerical calculations, the on-lattice velocity field,  $\mathbf{v}(\mathbf{r}, t + \Delta t)$ , is evaluated from the physical quantities at time  $t$  by eqn (18), and then we move particles by  $\mathbf{r}_i(t + \Delta t) = \mathbf{r}_i(t) + \Delta t \mathbf{V}_i(t + \Delta t)$  off-lattice, where  $\Delta t$  is the time increment of the numerical integration. In our simulation, the units of length  $\ell$  and time  $\tau$  are related as  $\eta \ell / \rho \ell^2 = 1$ , which sets both the scaled density and viscosity of the fluid region to unity. The units of stress and energy are  $\bar{\sigma} = \rho(\ell/\tau)^2$  and  $\bar{\epsilon} = \bar{\sigma} \ell^3$ , respectively.  $L_0$  is scaled as  $L_0 r_0 \tau / \ell^2 = 1$ . In the free energy functional, we set  $r_0 / \sigma = m / r_0 = K_0 / \ell^2 r_0 = 1$ ,  $\gamma / \ell r_0 = -10$ , and  $\chi / r_0 = 1$ . Furthermore, we set  $\eta_c / \eta_\ell = 50$ ,  $\Delta t = 0.003$ ,  $\xi = 1$ , and the average concentration to 0. To avoid cumbersome expressions, we will use the same characters for the scaled variables in the following. We solve the equation of motion, eqn (18), using the Marker-and-Cell (MAC) method with a staggered lattice under periodic boundary conditions.

The thermal fluctuation effects are neglected in the present analysis, which is thus at the mean-field level.

To check the validity of our simulation method, we examine both the static and dynamic behaviors for a single particle case by comparing FPD results with those of the sharp-interface case. Let us consider a particle ( $a = 6.4$ ) immersed in a critical binary mixture. In a quiescent state (without the average velocity field), the system is spherically symmetric around the center of mass of the particle at  $r = 0$ . In the sharp boundary case, we employ the method proposed in ref. 48. Fig. 5 shows the equilibrium concentration profile  $\psi(r)$  around a particle for various  $\varepsilon$  in both the sharp and smooth boundary cases. The adsorbed layer with  $\psi > 0$ , which is due to the negative value of  $\gamma (= -10)$ , is formed around the particle. The characteristic thickness of the adsorbed layer increases while approaching  $T_c$ , and it is the order of the bulk correlation length of the composition fluctuations ( $\sim (r_0\varepsilon/K_0)^{-1/2}$ ). We can see that, at each  $\varepsilon$ ,  $\psi(r)$  of both cases collapse onto a single curve besides the regions very near the surface ( $(r - a) \gtrsim \xi$ ).

We also calculate the drag force  $F_d$  acting on a particle moving with a constant velocity  $v_0$ : for the sharp boundary case, to calculate  $F_d$ , we consider a system where a fluid is flowing around a particle at rest with a velocity  $v_0$  at an infinite distance from the particle, instead of moving the particle. The stick boundary condition  $\mathbf{v} = 0$  at  $r = a$  is applied in the sharp boundary case. Fig. 6 shows the drag force  $F_d$  on a particle as a function of  $v_0$  in both the sharp and smooth boundary cases. When the particle surface is not coupled with concentration fluctuations, the Stokes-drag law  $F_d = \lambda^T 6\pi\eta_0 a v_0$  is reproduced with high accuracy. On the other hand, when there is a coupling between the particle surface and concentration fluctuations,  $F_d$  becomes larger by a factor of 20–30%. This extra dissipation for a particle coated with the adsorbed layer is already a manifestation of nontrivial nonequilibrium effects. From these static and dynamic behaviors of a single particle system, we confirm that our simulation (FPD) method with the above scheme well approximates the result of the sharp interface case.



**Fig. 6** The drag force plotted as a function of  $v_0$  for both sharp and smooth boundary cases. We can see that when a particle is suspended in a single-component liquid, the Stokes drag law  $F_d = \lambda^T 6\pi\eta_0 a v_0$  (green dot dashed line) is reproduced with high accuracy in the range of  $v_0$  studied. On the other hand, when a particle is immersed in a critical binary mixture and accompanied by concentration fluctuations, the deviation from the Stokes law becomes evident, where  $F_d$  becomes larger by a factor of 20–30%.

## 4 Summary

In this paper, we consider the physical basis of the FPD method to incorporate (i) thermal noise and (ii) local coupling of the colloid surface field with an order parameter of a fluid component. For problem (i), we introduce thermal stress noise into a fluid, which is a physically natural manner for the FPD method that treats a colloidal particle as a very viscous fluid particle. We confirm the validity of this method by analyzing the thermal Brownian motion of a single particle and the velocity correlation in the fluid. For problem (ii), we incorporate the coupling of the colloidal surface with the internal degrees of freedom of a fluid surrounding colloidal particles. Because of the smooth interface approximation, the treatment of such a coupling localized on the colloidal surface needs special care. We establish the theoretical basis to deal with such couplings in the FPD method and confirm its validity by comparing the FPD results with the results for a sharp-interface limit. This should allow us to study the dynamical behavior of colloidal particles suspended in a critical binary mixture, a nematic liquid crystal, and a quantum liquid He, where the coupling between the surface field of colloids and the order parameter field in a liquid component plays a crucial role.

The FPD method has significant merits in simulating the dynamical behavior of colloidal suspensions. It is based on the well-established Navier–Stokes equation with nonlocal viscosity. The physical assumptions are transparent and the physical principle behind the method is simple and very easy to understand. We can impose the incompressible condition, which plays a key role in many-body hydrodynamic interactions between colloids, in a straightforward manner. Our method can also include nonlinear inertia effects and thus be used with a relatively high Reynolds number (see, e.g., ref. 49). We hope that the FPD method will contribute to the physical understanding of hydrodynamic effects on the dynamical behavior of colloids suspended in simple and complex fluids as well as active colloids. We can also make a polymer or protein by connecting the fluid particles with springs and study its dynamical properties with hydrodynamic effects.<sup>5,7</sup>

## Conflicts of interest

There are no conflicts to declare.

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## 一、泛函形式

假设你有一个泛函  $F[\psi]$ ，它可以是如下形式：

$$F[\psi] = \int d\mathbf{r} L(\psi, \nabla\psi, \mathbf{r})$$

其中， $L$  是拉格朗日量（或汉密尔顿量），它依赖于场  $\psi$ 、场的梯度  $\nabla\psi$  和位置  $\mathbf{r}$ 。

## 二、泛函导数的基本定义

泛函导数的定义是：给定一个泛函  $F[\psi]$ ，其对  $\psi(\mathbf{r})$  的泛函导数定义为：

$$\frac{\delta F[\psi]}{\delta\psi(\mathbf{r})} = \lim_{\epsilon \rightarrow 0} \frac{F[\psi + \epsilon\delta\psi] - F[\psi]}{\epsilon}$$

这个表达式描述了泛函  $F[\psi]$  对于  $\psi(\mathbf{r})$  微小变化  $\delta\psi(\mathbf{r})$  的响应。

## 三、计算泛函导数的步骤

### 1. 确定泛函的形式

假设你已经有了一个具体的泛函，例如：

$$F[\psi] = \int d\mathbf{r} [f(\psi, \nabla\psi, \mathbf{r})]$$

其中  $f(\psi, \nabla\psi, \mathbf{r})$  是某个已知的函数。

### 2. 对 $\psi$ 的变化进行泰勒展开

对泛函进行泰勒展开，即：

$$F[\psi + \epsilon\delta\psi] = \int d\mathbf{r} [f(\psi + \epsilon\delta\psi, \nabla(\psi + \epsilon\delta\psi), \mathbf{r})]$$

展开到一阶的形式：

$$F[\psi + \epsilon\delta\psi] \approx F[\psi] + \epsilon \int d\mathbf{r} \left[ \frac{\partial f}{\partial\psi} \delta\psi + \frac{\partial f}{\partial(\nabla\psi)} \cdot \nabla\delta\psi \right]$$

### 3. 计算变分

泛函的导数可以通过以下方式计算：

$$\frac{\delta F[\psi]}{\delta \psi(\mathbf{r})} = \frac{\partial f}{\partial \psi} - \nabla \cdot \left( \frac{\partial f}{\partial (\nabla \psi)} \right)$$

其中：

- $\frac{\partial f}{\partial \psi}$  是拉格朗日量对  $\psi$  的偏导数，
- $\frac{\partial f}{\partial (\nabla \psi)}$  是拉格朗日量对梯度项的偏导数，
- $\nabla \cdot \left( \frac{\partial f}{\partial (\nabla \psi)} \right)$  是梯度部分的散度。

## 四、具体例子

假设你有一个泛函：

$$F[\psi] = \int d\mathbf{r} \left( \frac{1}{2} (\nabla \psi)^2 + \frac{1}{4} u_0 \psi^4 \right)$$

其中，第一项是梯度平方项，第二项是  $\psi^4$  项。

1. 计算对  $\psi$  的变分：

$$\frac{\partial}{\partial \psi} \left( \frac{1}{2} (\nabla \psi)^2 \right) = 0 \quad (\text{该项不依赖于 } \psi)$$

$$\frac{\partial}{\partial \psi} \left( \frac{1}{4} u_0 \psi^4 \right) = u_0 \psi^3$$

2. 计算梯度部分的变分：

对于  $\frac{1}{2} (\nabla \psi)^2$  项，我们有：

$$\frac{\partial}{\partial (\nabla \psi)} \left( \frac{1}{2} (\nabla \psi)^2 \right) = \nabla \psi$$

然后计算散度：

$$\nabla \cdot (\nabla \psi) = \nabla^2 \psi$$

### 3. 最终结果：

将这些结果代入欧拉-拉格朗日方程：

$$\frac{\delta F[\psi]}{\delta \psi(\mathbf{r})} = u_0 \psi^3 - \nabla^2 \psi$$

这就是给定泛函的泛函导数。

假设我们有一个二维热传导方程：

$$\frac{\partial u}{\partial t} = \alpha \nabla^2 u$$

这里  $u(x, y, t)$  是温度场， $\alpha$  是热扩散系数， $\nabla^2 u$  是空间的二阶导数。

- **时间导数：**使用前向差分（对时间离散化）：

$$\frac{\partial u}{\partial t} \approx \frac{u(x, y, t + \Delta t) - u(x, y, t)}{\Delta t}$$

- **空间导数：**使用中心差分（对空间离散化）：

$$\nabla^2 u \approx \frac{u(x + \Delta x, y) - 2u(x, y) + u(x - \Delta x, y)}{(\Delta x)^2} + \frac{u(x, y + \Delta y) - 2u(x, y) + u(x, y - \Delta y)}{(\Delta y)^2}$$

#### 2. 构建差分格式

将上述差分表达式代入原始方程，我们得到离散化后的方程：

$$\frac{u(x, y, t + \Delta t) - u(x, y, t)}{\Delta t} = \alpha \left( \frac{u(x + \Delta x, y) - 2u(x, y) + u(x - \Delta x, y)}{(\Delta x)^2} + \frac{u(x, y + \Delta y) - 2u(x, y) + u(x, y - \Delta y)}{(\Delta y)^2} \right)$$

假设积分范围不依赖于  $r_i$ ，并且函数  $|\nabla \phi_i|$  和  $f_s(\psi, T)$  在积分范围内是足够光滑的，那么你可以交换积分和求导的顺序：

$$\frac{\partial F_s}{\partial r_i} = \sum_i^N \int dr \frac{\partial}{\partial r_i} (|\nabla \phi_i| f_s(\psi, T)).$$

#### 注意事项

1. **连续性：**确保被积函数  $|\nabla \phi_i| f_s(\psi, T)$  在积分范围内是连续的。
2. **可微性：**确保  $|\nabla \phi_i|$  和  $f_s(\psi, T)$  对  $r_i$  是可微的。
3. **积分限：**如果积分限依赖于  $r_i$ ，则需要使用完整的莱布尼茨规则。

如果这些条件都满足，那么交换积分和求导的顺序是合理的。如果你有具体的函数形式或更多的上下文信

#### 5. Final expression:

- Assuming  $f_s(\psi, T)$  does not depend on  $r_i$ , the derivative simplifies to:

$$\frac{\partial F_s}{\partial r_i} = \sum_i^N \int dr f_s(\psi, T) \frac{\partial |\nabla \phi_i|}{\partial r_i}.$$

## 步骤 1: 简化表达式

首先, 我们可以简化  $\phi_i(r)$  的表达式:

$$\phi_i(r) = \frac{1}{2} \tanh \left[ \frac{1}{\xi_i} (a_i - |r - r_i|) \right] + \frac{1}{2}.$$

## 步骤 2: 求导

我们需要计算  $\nabla \phi_i(r)$ , 即  $\phi_i(r)$  对  $r$  的梯度。梯度是一个向量, 其分量是  $\phi_i(r)$  对各个空间坐标的偏导数。

### 2.1 计算 $\tanh$ 函数的导数

首先, 我们计算  $\tanh$  函数的导数:

$$\frac{d}{dx} \tanh(x) = \text{sech}^2(x) = 1 - \tanh^2(x).$$

$$\nabla \phi_i(r) = -\frac{1}{2\xi_i} \cdot \text{sech}^2 \left( \frac{1}{\xi_i} (a_i - |r - r_i|) \right) \cdot \frac{r - r_i}{|r - r_i|}.$$

要求梯度的模  $|\nabla \phi_i(r)|$  对  $r_i$  的导数, 我们需要对  $|\nabla \phi_i(r)|$  关于  $r_i$  进行求导。我们已经得到了  $|\nabla \phi_i(r)|$  的表达式:

$$|\nabla \phi_i(r)| = \frac{1}{2\xi_i} \cdot \text{sech}^2 \left( \frac{1}{\xi_i} (a_i - |r - r_i|) \right).$$

### 步骤 1: 简化表达式

首先, 我们可以将  $|\nabla \phi_i(r)|$  表示为:

$$|\nabla \phi_i(r)| = \frac{1}{2\xi_i} \cdot \text{sech}^2(u),$$

其中:

$$u = \frac{1}{\xi_i} (a_i - |r - r_i|).$$

## 步骤 2：求导

我们需要计算  $\frac{\partial}{\partial r_i} |\nabla \phi_i(r)|$ 。根据链式法则：

$$\frac{\partial}{\partial r_i} |\nabla \phi_i(r)| = \frac{1}{2\xi_i} \cdot \frac{\partial}{\partial r_i} [\text{sech}^2(u)] .$$

### 2.1 计算 $\frac{\partial}{\partial r_i} \text{sech}^2(u)$

首先，计算  $\text{sech}^2(u)$  对  $r_i$  的导数：

$$\frac{\partial}{\partial r_i} \text{sech}^2(u) = 2\text{sech}(u) \cdot (-\text{sech}(u) \tanh(u)) \cdot \frac{\partial u}{\partial r_i} .$$

由于  $\text{sech}(u) \cdot (-\text{sech}(u) \tanh(u)) = -\text{sech}^2(u) \tanh(u)$ ，所以：

$$\frac{\partial}{\partial r_i} \text{sech}^2(u) = -2\text{sech}^2(u) \tanh(u) \cdot \frac{\partial u}{\partial r_i} .$$

### 2.2 计算 $\frac{\partial u}{\partial r_i}$

接下来，计算  $u$  对  $r_i$  的导数：

$$u = \frac{1}{\xi_i} (a_i - |r - r_i|) .$$

因此：

$$\frac{\partial u}{\partial r_i} = \frac{1}{\xi_i} \cdot \frac{\partial}{\partial r_i} (a_i - |r - r_i|) = -\frac{1}{\xi_i} \cdot \frac{\partial}{\partial r_i} |r - r_i| .$$

我们知道：

$$\frac{\partial}{\partial r_i} |r - r_i| = -\frac{r - r_i}{|r - r_i|} .$$

因此：

$$\frac{\partial u}{\partial r_i} = -\frac{1}{\xi_i} \cdot \left( -\frac{r - r_i}{|r - r_i|} \right) = \frac{1}{\xi_i} \cdot \frac{r - r_i}{|r - r_i|} .$$

### 2.3 组合结果

将上述结果代入  $\frac{\partial}{\partial r_i} |\nabla \phi_i(r)|$  的表达式：

$$\frac{\partial}{\partial r_i} |\nabla \phi_i(r)| = \frac{1}{2\xi_i} \cdot \left( -2\text{sech}^2(u) \tanh(u) \cdot \frac{1}{\xi_i} \cdot \frac{r - r_i}{|r - r_i|} \right) .$$

简化后得到：

$$\frac{\partial}{\partial r_i} |\nabla \phi_i(r)| = -\frac{1}{\xi_i^2} \cdot \text{sech}^2(u) \tanh(u) \cdot \frac{r - r_i}{|r - r_i|} .$$

### 最终表达式

因此，梯度的模  $|\nabla \phi_i(r)|$  对  $r_i$  的导数为：

$$\frac{\partial}{\partial r_i} |\nabla \phi_i(r)| = -\frac{1}{\xi_i^2} \cdot \text{sech}^2 \left( \frac{1}{\xi_i} (a_i - |r - r_i|) \right) \tanh \left( \frac{1}{\xi_i} (a_i - |r - r_i|) \right) \cdot \frac{r - r_i}{|r - r_i|} .$$

这个表达式给出了梯度的模对  $r_i$  的导数，它取决于距离  $|r - r_i|$  和参数  $\xi_i$  和  $a_i$ 。

$$\frac{\partial \phi_i(r)}{\partial r_i} = \frac{1}{2\xi_i} \cdot \operatorname{sech}^2 \left( \frac{1}{\xi_i} (a_i - |r - r_i|) \right) \cdot \frac{r - r_i}{|r - r_i|}.$$

$$\operatorname{sech}^2(x) = 1 - \tanh^2(x)$$

## 1. $\phi$ as the Actual Composition Difference

If  $\phi$  represents the **actual composition difference** (e.g.,  $\phi = c_A - c_B$  in a binary mixture), and you are studying an **off-critical mixture**, you typically need to include a **cubic term** in the free energy density  $f(\phi)$  to account for the asymmetry of the system. The functional takes the form:

$$F[\phi] = \int \left[ \frac{1}{2} \kappa (\nabla \phi)^2 + f(\phi) \right] d^d r$$

with the local free energy density:

$$f(\phi) = \frac{a}{2} \phi^2 + \frac{b}{3} \phi^3 + \frac{c}{4} \phi^4$$

- **Cubic Term ( $\frac{b}{3} \phi^3$ ):** This term breaks the symmetry of the free energy landscape and reflects the fact that the two phases in an off-critical mixture have different compositions and free energies.
- **Quadratic ( $\frac{a}{2} \phi^2$ ) and Quartic ( $\frac{c}{4} \phi^4$ ) Terms:** These terms ensure the stability of the system and describe the energy cost of deviations from the equilibrium compositions.

## 2. $\phi$ as the Relative Difference (Deviation from Average)

If  $\phi$  represents the **relative difference** (i.e., the deviation from the average composition,  $\phi = c - c_{\text{avg}}$ ), the free energy functional can often be written using only **quadratic and quartic terms** in  $f(\phi)$ , even for an off-critical mixture. The functional takes the form:

$$F[\phi] = \int \left[ \frac{1}{2} \kappa (\nabla \phi)^2 + f(\phi) \right] d^d r$$

with the local free energy density:

$$f(\phi) = \frac{a}{2} \phi^2 + \frac{c}{4} \phi^4$$



- **No Cubic Term:** When  $\phi$  is defined as the deviation from the average composition, the asymmetry of the off-critical mixture is effectively "absorbed" into the definition of  $\phi$ . The free energy landscape is symmetric around  $\phi = 0$ , and the cubic term is not needed.
- **Quadratic ( $\frac{a}{2} \phi^2$ ) and Quartic ( $\frac{c}{4} \phi^4$ ) Terms:** These terms describe the energy cost of fluctuations around the average composition and stabilize the system.

## Why the Difference?

- **Actual Composition Difference:** When  $\phi$  is the actual composition difference, the free energy landscape must explicitly account for the asymmetry of the off-critical mixture. This is why a cubic term is needed.
- **Relative Difference:** When  $\phi$  is the deviation from the average composition, the asymmetry is implicitly included in the definition of  $\phi$ , so the free energy landscape can be symmetric, and a cubic term is unnecessary.

### Summary

- If  $\phi$  is the **actual composition difference**, include a **cubic term** in  $f(\phi)$  to describe an off-critical mixture.
- If  $\phi$  is the **relative difference (deviation from average)**, you can use only **quadratic and quartic terms** in  $f(\phi)$ , even for an off-critical mixture.

Both approaches are valid, but the choice depends on the physical interpretation of  $\phi$  and the specific problem you are studying.

### Summary

- **Off-critical** refers to the **asymmetry in the phase diagram**, not necessarily the initial composition.
- Even if the initial amounts of A and B are equal (e.g., 50-50), the system can still be off-critical if the phase diagram is asymmetric.
- In an off-critical system, phase separation results in two phases with **asymmetric compositions**, reflecting the asymmetry of the phase diagram.
- The **critical composition** is a property of the phase diagram and represents the composition at which critical behavior occurs.
- The **initial amounts of A and B** in a system can be equal to or different from the critical composition.
- If the initial composition is **off-critical** (different from the critical composition), the system will exhibit asymmetric phase separation, even if the phase diagram itself is symmetric.

## 1. Phase Diagram:

- The phase diagram is a **thermodynamic construct** that shows the equilibrium phases of a system as a function of variables like temperature, pressure, and composition.
- It is determined by the underlying free energy landscape of the system and does not depend on the initial conditions (e.g., initial composition).

## 2. Off-Critical:

- The term **off-critical** refers to the **location of the system** on the phase diagram relative to the critical point. It describes whether the system's composition (or other variables) is different from the critical composition (or critical temperature/pressure).
- Off-critical is a **descriptor of the system's state** within the phase diagram, not a property of the phase diagram itself.

